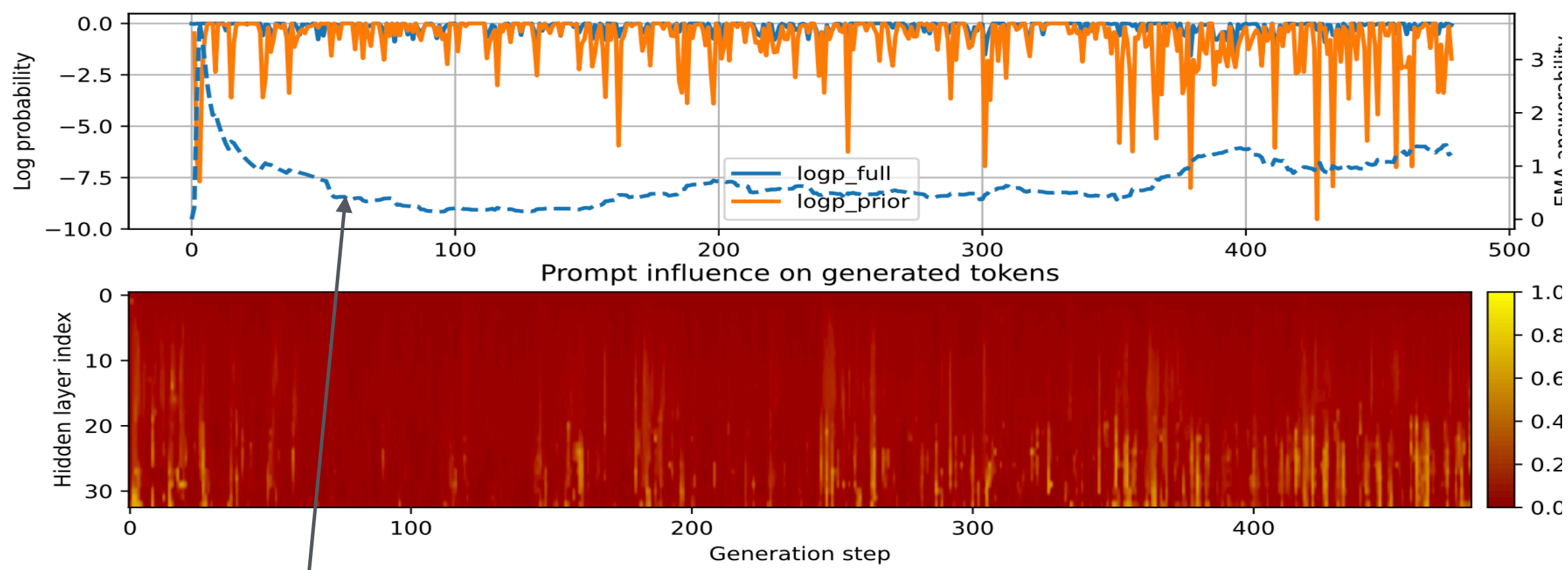
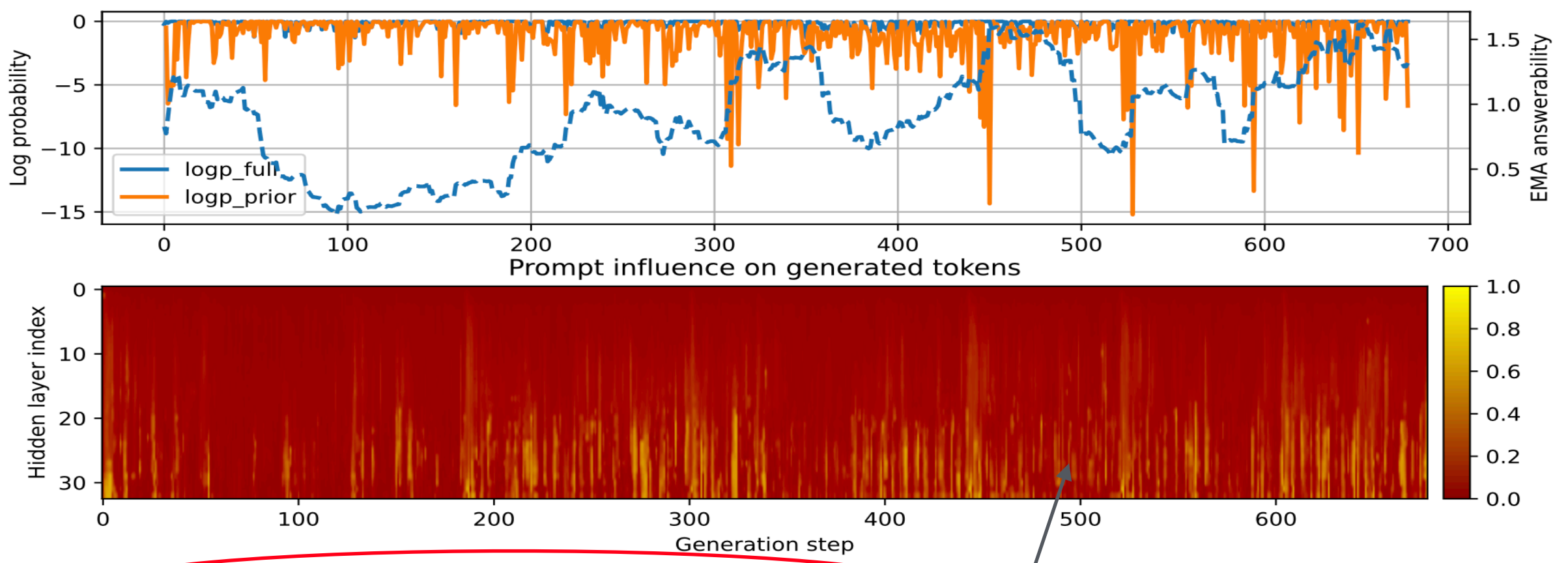




PROMPT:
Explain the difference between KL divergence and Jensen-Shannon divergence. Include when each one is preferable in machine learning applications.

MODEL RESPONSE:
Kullback-Leibler (KL) divergence and Jensen-Shannon (JS) divergence are both measures of the difference between two probability distributions. However, they have different properties and are used in different contexts. 1. Kullback-Leibler Divergence (KL Divergence): KL divergence measures the amount of information lost when approximating one distribution with another. It is not symmetric, meaning $KL(P||Q) \neq KL(Q||P)$, and it is defined as: $KL(P||Q) = \sum P(x) \log(P(x)/Q(x))$ where P and Q are the two probability distributions. KL divergence is undefined when $P(x) = 0$ for any x where $Q(x) > 0$. In machine learning applications, KL divergence is often used in generative models, such as variational autoencoders (VAEs), to measure the distance between the learned distribution and the true data distribution. It is also used in reinforcement learning to compare the current policy with the optimal policy. 2. Jensen-Shannon Divergence (JS Divergence): JS divergence is a symmetric measure of the difference between two probability distributions. It is defined as the average of two KL divergences, each computed with respect to a mixture distribution: $JS(P||Q) = 1/2 * [KL(P||M) + KL(Q||M)]$ where $M = (P + Q) / 2$ is the mixture distribution. JS divergence is always finite and non-negative. In machine learning applications, JS divergence is often used when dealing with multiple distributions, such as in clustering algorithms or when comparing the distributions of different features in a dataset. It is also used in information theory to measure the uncertainty of a probability distribution. In summary, KL divergence is preferable when comparing a single approximation to a true distribution, such as in generative models or reinforcement learning. JS divergence is preferable when dealing with multiple distributions or when symmetry is important, such as in clustering algorithms or information theory.



PROMPT:
Since Jensen-Shannon divergence is always larger than KL divergence and is unbounded, explain why it is usually preferred for detecting hallucinations in LLM outputs.

MODEL RESPONSE:
Jensen-Shannon (JS) divergence and Kullback-Leibler (KL) divergence are both measures of the difference between two probability distributions. However, they have different properties that make them suitable for different applications. 1. KL Divergence: KL divergence is asymmetric, meaning it is not equal in both directions. It measures the amount of information lost when approximating one distribution with another. It is defined as the sum of the product of the probabilities of each event and the logarithm of the ratio of the probabilities of that event in the two distributions. KL divergence is always non-negative and is zero if and only if the two distributions are identical. However, KL divergence is not defined when the probability of an event in one distribution is zero and the probability of that event in the other distribution is non-zero. 2. JS Divergence: JS divergence is a symmetric version of KL divergence, which means it is equal in both directions. It is the average of two KL divergences, one calculated from each distribution with respect to their common average distribution. JS divergence is always non-negative and is zero if and only if the two distributions are identical. Unlike KL divergence, JS divergence is defined even when the probability of an event in one distribution is zero and the probability of that event in the other distribution is non-zero. In the context of detecting hallucinations in Language Model (LM) outputs, JS divergence is usually preferred over KL divergence for several reasons: 1. Boundedness: JS divergence is bounded, which means it has a maximum value. This is important because in the case of hallucinations, the LM might generate output that is not in the training data, and the probability of such events can be very small or even zero. If we use KL divergence in this case, it would be undefined or very large, making it difficult to compare the divergence between different LMs or to set a threshold for what constitutes a hallucination. 2. Symmetry: JS divergence

$$\log \left(\frac{p(nextToken | tokenHistory, prompt)}{p(nextToken | tokenHistory)} \right)$$

False premise injection

Layer-wise distribution divergence (spatio-temporal behavior)

- Low value = Auto-regressive trajectory (low prompt influence).
- High value = High prompt influence on trajectory.

- Layerwise hidden states projected into token-logit space.
- Distribution divergence between $p(nextToken|tokenHistory, prompt)$ and $p(nextToken|tokenHistory)$ is measured.
- Red = Low prompt influence (self-sustaining).
- Yellow = high prompt influence (periodic re-grounding)